Nitrogen is needed for the synthesis of amino acids, nucleotides like purines and pyrimidines which are part of nucleic acids (DNA and RNA).

Phosphorus is a part of phospholipids, nucleotides like ATP and phosphodiester bonds of nucleic acids.

Carbon, Hydrogen and Oxygen are the backbone of all organic macromolecules like peptidoglycan, proteins and lipids and nucleic acids.

Sulphur is needed for the synthesis of thiamin, biotin, and aminoacids like cysteine and methionine.

Potassium, Calcium, Magnesium and Iron exist as cations in the cell. These element plays vital role in the metabolic activity of microorganisms. Potassium (K^+) is needed for the activity of many enzymes Example: Pyruvate Kinase.

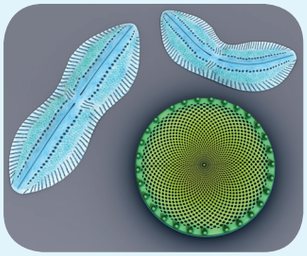
Calcium (Ca^{2+}) is involved in the heat resistance of bacterial endospores.

Magnesium (Mg^{2+}) binds with ATP and serves as a cofactor of enzymes like hexokinase.

Iron (Fe^{2+} or Fe^{3+}) is present in cytochromes and act as cofactors for cytochrome oxidase, catalase and peroxidase.



Diatoms (A group of algae) need silicon to construct their beautiful cell walls.



Micronutrients

Nutrients that are needed in trace quantities are called micronutrients. Example: Zinc (Zn), Molybdenum (Mo), Cobalt (Co), Manganese (Mn).

Besides macro and micronutrients, some microorganisms need growth factors like amino acids, purines and pyrimidines and vitamins. Example: Biotin is required by *Leuconostoc* sp and folic acid is required by *Enterococcus faecalis*.

HOTS

Is there a microbe that can grow in a medium that contains only the following compounds in water: calcium carbonate, magnesium nitrate, ferrous chloride, zinc sulphate and glucose. Defend your answer.

6.3 Nutritional Types of Microorganisms

Microorganisms can be classified into nutritional classes based on how they satisfy the requirements of carbon, energy and electrons for their growth and nutrition.

Based on the carbon source, microorganisms are able to utilize, they are classified into Autotrophs and Heterotrophs.

Autotrophs: These are organisms that utilize CO_2 as their sole source of carbon.

Heterotrophs: These are organisms that use preformed organic substances from other organisms as their carbon source.

Based on energy source, microorganisms are classified into Phototrophs and Chemotrophs.

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